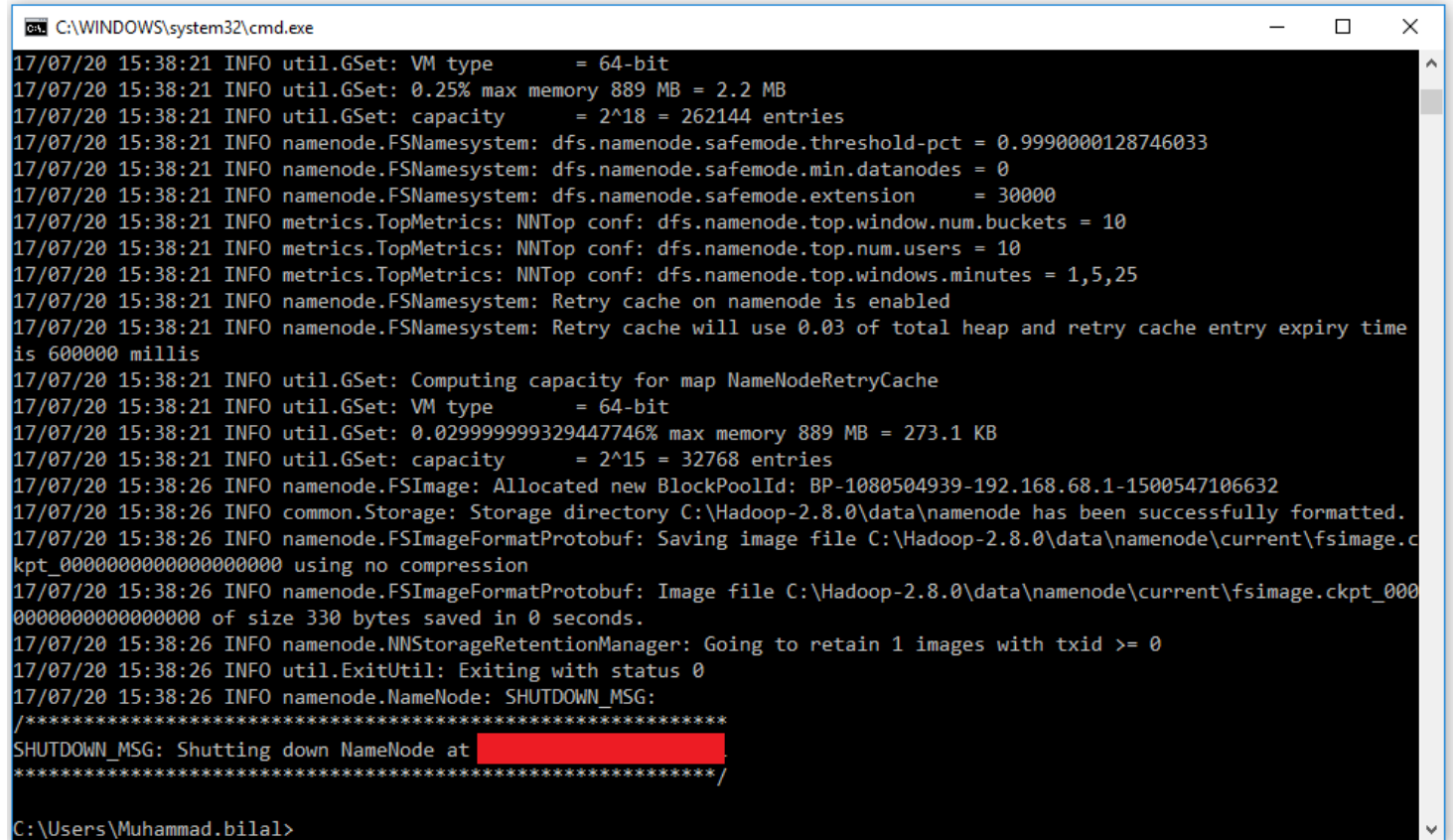


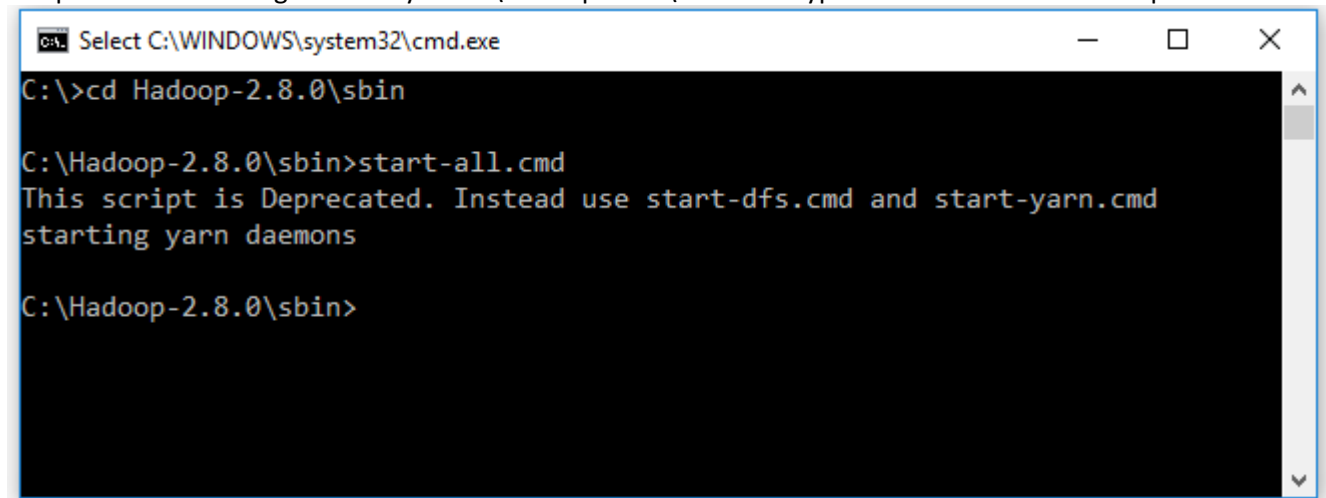
1. Open cmd and typing command "**hdfs namenode -format**". You will see



```
C:\WINDOWS\system32\cmd.exe
17/07/20 15:38:21 INFO util.GSet: VM type      = 64-bit
17/07/20 15:38:21 INFO util.GSet: 0.25% max memory 889 MB = 2.2 MB
17/07/20 15:38:21 INFO util.GSet: capacity    = 2^18 = 262144 entries
17/07/20 15:38:21 INFO namenode.FSNamesystem: dfs.namenode.safemode.threshold-pct = 0.9990000128746033
17/07/20 15:38:21 INFO namenode.FSNamesystem: dfs.namenode.safemode.min.datanodes = 0
17/07/20 15:38:21 INFO namenode.FSNamesystem: dfs.namenode.safemode.extension  = 30000
17/07/20 15:38:21 INFO metrics.TopMetrics: NNTop conf: dfs.namenode.top.window.num.buckets = 10
17/07/20 15:38:21 INFO metrics.TopMetrics: NNTop conf: dfs.namenode.top.num.users = 10
17/07/20 15:38:21 INFO metrics.TopMetrics: NNTop conf: dfs.namenode.top.windows.minutes = 1,5,25
17/07/20 15:38:21 INFO namenode.FSNamesystem: Retry cache on namenode is enabled
17/07/20 15:38:21 INFO namenode.FSNamesystem: Retry cache will use 0.03 of total heap and retry cache entry expiry time
is 600000 millis
17/07/20 15:38:21 INFO util.GSet: Computing capacity for map NameNodeRetryCache
17/07/20 15:38:21 INFO util.GSet: VM type      = 64-bit
17/07/20 15:38:21 INFO util.GSet: 0.029999999329447746% max memory 889 MB = 273.1 KB
17/07/20 15:38:21 INFO util.GSet: capacity    = 2^15 = 32768 entries
17/07/20 15:38:26 INFO namenode.FSImage: Allocated new BlockPoolId: BP-1080504939-192.168.68.1-1500547106632
17/07/20 15:38:26 INFO common.Storage: Storage directory C:\Hadoop-2.8.0\data\namenode has been successfully formatted.
17/07/20 15:38:26 INFO namenode.FSImageFormatProtobuf: Saving image file C:\Hadoop-2.8.0\data\namenode\current\fsimage.ck
kpt_000000000000000000 using no compression
17/07/20 15:38:26 INFO namenode.FSImageFormatProtobuf: Image file C:\Hadoop-2.8.0\data\namenode\current\fsimage.ckpt_000
00000000000000000 of size 330 bytes saved in 0 seconds.
17/07/20 15:38:26 INFO namenode.NNStorageRetentionManager: Going to retain 1 images with txid >= 0
17/07/20 15:38:26 INFO util.ExitUtil: Exiting with status 0
17/07/20 15:38:26 INFO namenode.NameNode: SHUTDOWN_MSG:
/*****
SHUTDOWN_MSG: Shutting down NameNode at [REDACTED]
*****/
C:\Users\Muhammad.bilal>
```

Testing

2. Open cmd and change directory to "C:\Hadoop-2.8.0\sbin" and type "**start-all.cmd**" to start apache.



```
Select C:\WINDOWS\system32\cmd.exe
C:\>cd Hadoop-2.8.0\sbin

C:\Hadoop-2.8.0\sbin>start-all.cmd
This script is Deprecated. Instead use start-dfs.cmd and start-yarn.cmd
starting yarn daemons

C:\Hadoop-2.8.0\sbin>
```

3. Make sure these apps are running

- Hadoop Namenode
- Hadoop datanode
- YARN Resourc Manager
- YARN Node Manager

```
17/07/20 15:50:09 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/20 15:50:12 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/20 15:50:15 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/20 15:50:18 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/20 15:50:21 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/20 15:50:24 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/20 15:50:27 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/20 15:50:30 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/20 15:50:33 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/20 15:50:36 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/20 15:50:39 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/20 15:50:42 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/20 15:50:46 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/20 15:50:49 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/20 15:50:52 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/20 15:50:55 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/20 15:50:58 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/20 15:51:01 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/20 15:51:04 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/20 15:51:07 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/20 15:51:10 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/20 15:51:13 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/20 15:51:16 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/20 15:51:19 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/20 15:51:22 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/20 15:51:25 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/20 15:51:29 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/20 15:51:32 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/20 15:51:35 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
```

4. Open: <http://localhost:8088>

← → ↻ 🌐 localhost:8088/cluster

🌟 ⋮

Logged in as: dr.who



All Applications

Cluster

About

Nodes

Node Labels

Applications

NEW

NEW SAVING

SUBMITTED

ACCEPTED

RUNNING

FINISHED

FAILED

KILLED

Scheduler

Tools

Cluster Metrics

Apps Submitted	Apps Pending	Apps Running	Apps Completed	Containers Running	Memory Used	Memory Total	Memory Reserved	VCores Used	VCores Total	VCores Reserved
0	0	0	0	0	0 B	8 GB	0 B	0	8	0

Cluster Nodes Metrics

Active Nodes	Decommissioning Nodes	Decommissioned Nodes	Lost Nodes	Unhealthy Nodes	Rebooted Nodes	Shutdown Nodes
1	0	0	0	0	0	0

Scheduler Metrics

Scheduler Type	Scheduling Resource Type	Minimum Allocation	Maximum Allocation	Maximum Cluster Application Priority
Capacity Scheduler	[MEMORY]	<memory:1024, vCores:1>	<memory:8192, vCores:4>	0

Show 20 entries

Search:

ID	User	Name	Application Type	Queue	Application Priority	StartTime	FinishTime	State	FinalStatus	Running Containers	Allocated CPU VCores	Allocated Memory MB	% of Queue	% of Cluster	Progress	Tracking UI	Blacklisted Nodes
No data available in table																	

Showing 0 to 0 of 0 entries

First Previous Next Last

5. Open: <http://localhost:50070>

localhost:50070/dfshealth.html#tab-overview

Hadoop

Overview

Datanodes

Datanode Volume Failures

Snapshot

Startup Progress

Utilities

Overview 'localhost:9000' (active)

Started:	Thu Jul 20 15:44:11 +0500 2017
Version:	2.8.0, r91f2b7a13d1e97b7cc29ac0009
Compiled:	Fri Mar 17 09:12:00 +0500 2017 by jdu from branch-2.8.0
Cluster ID:	CID-098b09fc-fc7b674
Block Pool ID:	BP-1080504947106632

Summary

Security is off.

Safemode is off.

1 files and directories, 0 blocks = 1 total filesystem object(s).

Heap Memory used 36.53 MB of 311 MB Heap Memory. Max Heap Memory is 889 MB.

Non Heap Memory used 40.68 MB of 41.53 MB Committed Non Heap Memory. Max Non Heap Memory is <unbounded>.

Configured Capacity:	475.24 GB
DFS Used:	321 B (0%)
Non DFS Used:	261.08 GB

Congratulations, Hadoop installed.

How to Run Hadoop wordcount MapReduce Example on Windows 10

Prepare

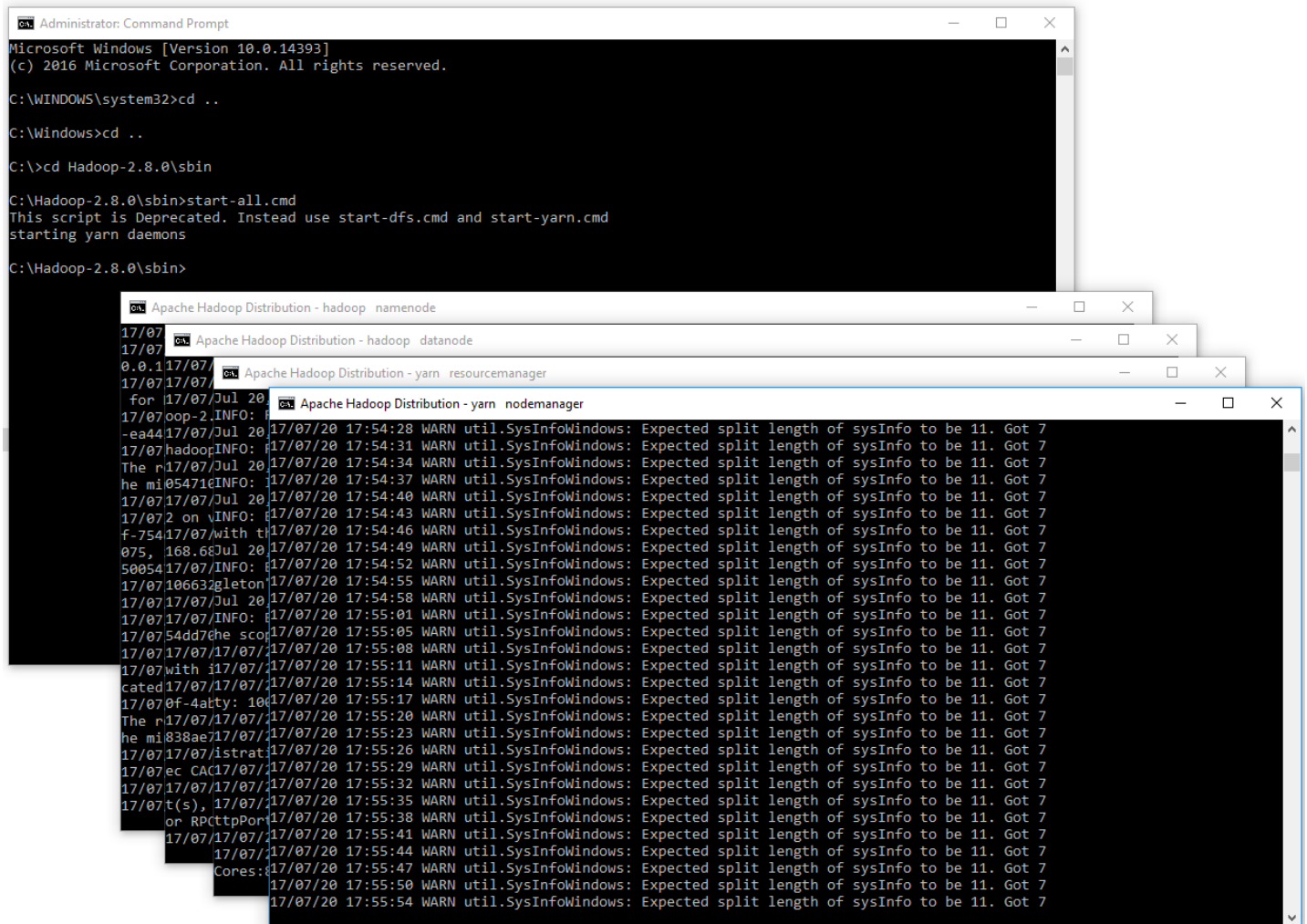
1. Download MapReduceClient.jar (Link: <https://github.com/MuhammadBilalYar/HADOOP-INSTALLATION-ON-WINDOW-10/blob/master/MapReduceClient.jar>)
2. Download Input_file.txt (Link: https://github.com/MuhammadBilalYar/HADOOP-INSTALLATION-ON-WINDOW-10/blob/master/input_file.txt)

Place both files in "C:/"

Hadoop Operation

1. Open cmd in Administrative mode and move to "C:/Hadoop-2.8.0/sbin" and start cluster

Start-all.cmd



The screenshot shows a Windows Command Prompt window titled "Administrator: Command Prompt" with the following commands and output:

```
Microsoft Windows [Version 10.0.14393]
(c) 2016 Microsoft Corporation. All rights reserved.

C:\WINDOWS\system32>cd ..
C:\Windows>cd ..
C:\>cd Hadoop-2.8.0\sbin
C:\Hadoop-2.8.0\sbin>start-all.cmd
This script is Deprecated. Instead use start-dfs.cmd and start-yarn.cmd
starting yarn daemons
C:\Hadoop-2.8.0\sbin>
```

Below the Command Prompt, several Hadoop service windows are visible, each showing log messages. The windows are titled:

- Apache Hadoop Distribution - hadoop namenode
- Apache Hadoop Distribution - hadoop datanode
- Apache Hadoop Distribution - yarn resourcemanager
- Apache Hadoop Distribution - yarn nodemanager

The log messages in the windows show various warnings and status updates, including:

```
17/07/20 17:54:28 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/20 17:54:31 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/20 17:54:34 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/20 17:54:37 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/20 17:54:40 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/20 17:54:43 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/20 17:54:46 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/20 17:54:49 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/20 17:54:52 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/20 17:54:55 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/20 17:54:58 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/20 17:55:01 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/20 17:55:05 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/20 17:55:08 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/20 17:55:11 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/20 17:55:14 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/20 17:55:17 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/20 17:55:20 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/20 17:55:23 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/20 17:55:26 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/20 17:55:29 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/20 17:55:32 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/20 17:55:35 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/20 17:55:38 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/20 17:55:41 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/20 17:55:44 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/20 17:55:47 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/20 17:55:50 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
17/07/20 17:55:54 WARN util.SysInfoWindows: Expected split length of sysInfo to be 11. Got 7
```

2. Create an input directory in HDFS.

```
hadoop fs -mkdir /input_dir
```

3. Copy the input text file named input_file.txt in the input directory (input_dir) of HDFS.

```
hadoop fs -put C:/input_file.txt /input_dir
```

4. Verify input_file.txt available in HDFS input directory (input_dir).

```
hadoop fs -ls /input_dir/
```

```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.14393]
(c) 2016 Microsoft Corporation. All rights reserved.

C:\WINDOWS\system32>cd/

C:\>cd Hadoop-2.8.0\sbin\

C:\Hadoop-2.8.0\sbin>start-all.cmd
This script is Deprecated. Instead use start-dfs.cmd and start-yarn.cmd
starting yarn daemons

C:\Hadoop-2.8.0\sbin>cd/

C:\>hadoop dfsadmin -safemode leave
DEPRECATED: Use of this script to execute hdfs command is deprecated.
Instead use the hdfs command for it.
Safe mode is OFF

C:\>hadoop fs -mkdir /input_dir

C:\>hadoop fs -put C:/input_file.txt /input_dir

C:\>hadoop fs -ls /input_dir/
Found 1 items
-rw-r--r--  1 Muhammad.Bilal supergroup      1888 2017-07-20 18:31 /input_dir/input_file.txt

C:\>
```

5. Verify content of the copied file.

```
hadoop dfs -cat /input_dir/input_file.txt
```

```

C:\>hadoop fs -ls /input_dir/
Found 1 items
-rw-r--r--    1 Muhammad.Bilal supergroup      1888 2017-07-20 18:31 /input_dir/input_file.txt

C:\>hadoop dfs -cat /input_dir/input_file.txt
DEPRECATED: Use of this script to execute hdfs command is deprecated.
Instead use the hdfs command for it.
23  23  27  43  24  25  26  26  26  26  25  26  25
26  27  28  28  28  30  31  31  31  30  30  30  29
31  32  32  32  33  34  35  36  36  34  34  34  34
39  38  39  39  39  41  42  43  40  39  38  38  40
38  39  39  39  39  41  41  41  28  40  39  39  45
23  23  27  43  24  25  26  26  26  26  25  26  25
26  27  28  28  28  30  31  31  31  30  30  30  29
31  32  32  32  33  34  35  36  36  34  34  34  34
39  38  39  39  39  41  42  43  40  39  38  38  40
38  39  39  39  39  41  41  41  28  40  39  39  45
23  23  27  43  24  25  26  26  26  26  25  26  25
26  27  28  28  28  30  31  31  31  30  30  30  29
31  32  32  32  33  34  35  36  36  34  34  34  34
39  38  39  39  39  41  42  43  40  39  38  38  40
38  39  39  39  39  41  41  41  28  40  39  39  45
23  23  27  43  24  25  26  26  26  26  25  26  25
26  27  28  28  28  30  31  31  31  30  30  30  29
31  32  32  32  33  34  35  36  36  34  34  34  34
39  38  39  39  39  41  42  43  40  39  38  38  40
38  39  39  39  39  41  41  41  28  40  39  39  45
23  23  27  43  24  25  26  26  26  26  25  26  25
26  27  28  28  28  30  31  31  31  30  30  30  29
31  32  32  32  33  34  35  36  36  34  34  34  34
39  38  39  39  39  41  42  43  40  39  38  38  40
38  39  39  39  39  41  41  41  28  40  39  39  45

```

6. Run MapReduceClient.jar and also provide input and out directories.

```
hadoop jar C:/MapReduceClient.jar wordcount /input_dir /output_dir
```

```
Administrator: Command Prompt
FILE: Number of large read operations=0
FILE: Number of write operations=0
HDFS: Number of bytes read=1999
HDFS: Number of bytes written=120
HDFS: Number of read operations=6
HDFS: Number of large read operations=0
HDFS: Number of write operations=2
Job Counters
  Launched map tasks=1
  Launched reduce tasks=1
  Data-local map tasks=1
  Total time spent by all maps in occupied slots (ms)=2180
  Total time spent by all reduces in occupied slots (ms)=2442
  Total time spent by all map tasks (ms)=2180
  Total time spent by all reduce tasks (ms)=2442
  Total vcore-milliseconds taken by all map tasks=2180
  Total vcore-milliseconds taken by all reduce tasks=2442
  Total megabyte-milliseconds taken by all map tasks=2232320
  Total megabyte-milliseconds taken by all reduce tasks=2500608
Map-Reduce Framework
  Map input records=30
  Map output records=390
  Map output bytes=2730
  Map output materialized bytes=195
  Input split bytes=111
  Combine input records=390
  Combine output records=21
  Reduce input groups=21
  Reduce shuffle bytes=195
  Reduce input records=21
  Reduce output records=21
  Spilled Records=42
  Shuffled Maps =1
  Failed Shuffles=0
  Merged Map outputs=1
  GC time elapsed (ms)=70
  CPU time spent (ms)=764
  Physical memory (bytes) snapshot=471478272
  Virtual memory (bytes) snapshot=619429888
  Total committed heap usage (bytes)=353894400
Shuffle Errors
  BAD_ID=0
  CONNECTION=0
  IO_ERROR=0
  WRONG_LENGTH=0
  WRONG_MAP=0
  WRONG_REDUCE=0
File Input Format Counters
  Bytes Read=1888
File Output Format Counters
  Bytes Written=120
c:\>
```

7. Verify content for generated output file.

```
hadoop dfs -cat /output_dir/*
```

```
C:\>hadoop dfs -cat /output_dir/*
DEPRECATED: Use of this script to execute hdfs command is deprecated.
Instead use the hdfs command for it.
23      12
24      6
25      18
26      36
27      12
28      24
29      6
30      24
31      24
32      18
33      6
34      30
35      6
36      12
38      24
39      66
40      18
41      24
42      6
43      12
45      6

C:\>
```

Some Other usefull commands

To leave Safe mode

```
hadoop dfsadmin -safemode leave
```

To Delete file from HDFS directory

```
hadoop fs -rm -r /input_dir/input_file.txt
```

To Delete directory from HDFS directory

```
hadoop fs -rm -r /input_dir
```

```
C:\>hadoop dfsadmin -safemode leave
DEPRECATED: Use of this script to execute hdfs command is deprecated.
Instead use the hdfs command for it.
Safe mode is OFF

C:\>hadoop fs -rm -r /input_dir/input_file.txt
Deleted /input_dir/input_file.txt

C:\>hadoop fs -rm -r /input_dir
Deleted /input_dir

C:\>
```